

Chronic lymphocytic leukemia / small lymphocytic lymphoma

CD5+ CD23+ B-cell neoplasm · **treat only on an iwCLL indication** · **TP53 + IGHV** pick between continuous BTKi and fixed-duration venetoclax · chemoimmunotherapy is obsolete for most

Compiled from iwCLL 2018 · NCCN · WHO-HEM5/ICC 2022 · primary trial reports verified against PubMed, October 2026.

DIAGNOSIS — CLL VS SLL VS MBL (IWCLL 2018, WHO-5)

Entity	Defining features
CLL	≥5×10 ⁹ /L clonal B cells in blood, sustained ≥3 mo ; clonality by light-chain restriction on flow. A cytopenia from typical marrow infiltrate = CLL at any count
SLL	Same disease, nodal: nodes and/or spleen, blood B cells <5×10⁹/L , no marrow-driven cytopenia; confirm on lymph node biopsy
MBL	Clonal B cells <5×10⁹/L , no nodes, organomegaly, cytopenia or symptoms. Low-count <0.5 (rarely progresses) vs high-count 0.5–4.9 . Progression to treatment-requiring CLL ~1–2%/yr; more infections and skin cancers
Prolymphocytic progression	WHO-5: >15% prolymphocytes in blood/marrow. WHO-5 dropped B-PLL (now CLL with PL progression, MCL variant, or splenic B-cell lymphoma with prominent nucleoli); ICC 2022 keeps B-PLL narrowly
Richter transformation	Aggressive lymphoma arising in CLL — DLBCL type (~90%) or Hodgkin type. Page 2

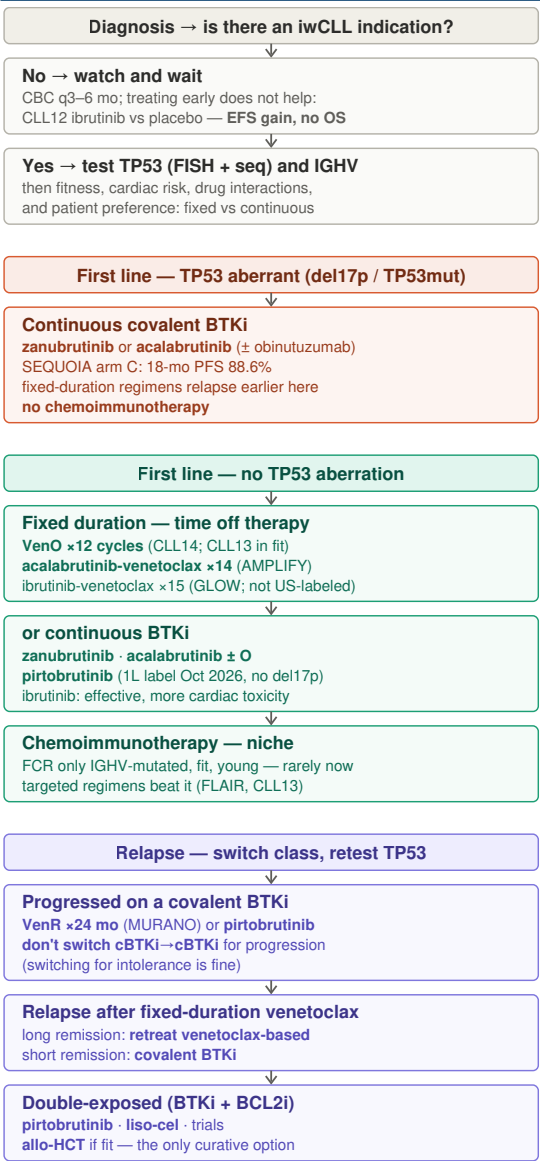
FLOW PHENOTYPE AND THE MIMICS

- **CD19+ CD5+ CD23+** with **dim CD20, dim CD79b, dim surface Ig** (single light chain). **CD200+**, LEF1+, ROR1+; FMC7–. Smudge cells on smear
- **Matutes (RMH) score**: 1 point each for CD5+, CD23+, FMC7–, weak slg, weak CD22 (CD79b in the modified score). **4–5 = CLL** (87% of CLL); 3 = atypical; 0–2 = other B-LPD
- **Mantle cell is the dangerous mimic** (CD5+): **CD23–, CD200–, bright slg/CD20, cyclin D1+, SOX11+, t(11;14)**. Atypical/CD23-dim cases: **cyclin D1 IHC or t(11;14) FISH**
- Others: marginal zone (CD5–, CD23–), hairy cell (CD11c/CD25/CD103, BRAF V600E), follicular (CD10+)

PRETREATMENT WORKUP (IWCLL)

- **Before every line of therapy: FISH** (del13q, del11q, +12, del17p) + **TP53 sequencing** — clones evolve; TP53 status flips the regimen
- **IGHV mutation status once** (doesn't change): **unmutated = ≥98% identity to germline**; mutated = >2% deviation
- **CpG-stimulated karyotype**: complex = ≥3 abnormalities; **≥5 is independently adverse**; 3–4 adverse mainly with TP53 aberration
- CBC + smear, CMP, LDH, uric acid, **β2-microglobulin, DAT** + retic/ haptoglobin, **quantitative IgG/IgA/IgM, HBV (HBsAg, anti-HBc), HCV, HIV**
- **CT neck/C/A/P before venetoclax** (TLS risk) or in trials; not needed for watch-and-wait. **PET only if Richter suspected**. Marrow not required for diagnosis — do it for unexplained cytopenias
- Cardiac history, ECG, BP, anticoagulants/antiplatelets and CYP3A drugs before choosing a BTKi

TREATMENT PATHWAY



IWCLL INDICATIONS TO TREAT — NEED ≥1

- **Progressive marrow failure**: Hb <10 g/dL or platelets <100×10⁹/L (a stable plt just under 100 may be watched)
- **Spleen ≥6 cm below costal margin**, or progressive/symptomatic · **nodes ≥10 cm**, or progressive/symptomatic
- **Lymphocyte doubling time <6 mo** or ≥50% rise in 2 mo (ALC <30: observe longer before calling LDT)
- **Autoimmune anemia/ITP poorly responsive to steroids** · symptomatic extranodal disease
- **Constitutional**: weight loss ≥10% in 6 mo · fatigue ECOG ≥2 · fever ≥38.0°C ≥2 wk · night sweats ≥1 mo (no infection)
- **NOT indications**: high ALC alone (leukostasis is rare), hypogammaglobulinemia, a paraprotein, adverse FISH/IGHV, Rai stage alone

STAGING

System	Definition · survival
Rai	0 lymphocytosis only (low) · I nodes · II spleen/liver (intermediate) · III Hb <11 · IV plt <100 (high). Original median survival >150 / 101 / 71 / 19 / 19 mo
Binet	Areas: head-neck, axillae, groins, spleen, liver. A ≤2 areas (~55%; near-normal survival) · B ≥3 areas (~30%) · C Hb <10 or plt <100 (~15%; median ~2 yr)

Anemia/thrombocytopenia from **autoimmune destruction does not upstage** — Rai III/IV and Binet C mean marrow failure.

CLL-IPi (2016) — PROGNOSIS, NOT A TREATMENT TRIGGER

Factor	Pts
TP53 aberrant (del17p and/or TP53 mutation)	4
IGHV unmutated · β2M >3.5 mg/L	2 each
Rai I–IV or Binet B–C · age >65	1 each

Low 0–1 · intermediate 2–3 · high 4–6 · very high 7–10 → 5-yr OS **93 / 79 / 63 / 23%** (chemo era; targeted agents narrow the gaps, esp. for IGHV).

GENETICS (DÖHNER HIERARCHY)

Lesion	Meaning
del(13q) sole ~36% (any ~55%)	Most favorable; median OS 133 mo (chemo era)
Trisomy 12 ~16%	Intermediate; often atypical (CD23-dim) phenotype
del(11q) ~18%	ATM; bulky nodes; adverse with chemo — largely overcome by BTKi
del(17p) / TP53 mut ~7% at dx; more at relapse	Chemo-refractory; never FCR/BR ; continuous BTKi preferred
IGHV unmutated ~40% at dx	Shorter remissions after fixed-duration and chemo; no plateau with FCR

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Page 2 — first-line evidence, relapsed disease and sequencing, Richter transformation

FIRST LINE — THE EVIDENCE

Regimen · trial	Population · result · the catch
Zanubrutinib · SEQUOIA	≥65 or comorbid, vs BR: 5-yr median PFS NR vs 44.1 mo, HR 0.29 (IGHV-U 0.21, IGHV-M 0.40); OS no different (85.8 vs 85.0%). Arm C del(17p) (single-arm, n=109): ORR 94.5%, 18-mo PFS 88.6%
Acalabrutinib ± O · ELEVATE-TN	Older/comorbid, vs Clb-O: 6-yr PFS 78% (A-O) / 62% (A) / 17% ; A-O vs A HR 0.58; OS HR 0.62 A-O vs Clb-O. The one BTKi trial where adding anti-CD20 clearly helps
Ibrutinib · RESONATE-2, A041202, E1912	RESONATE-2 (≥65): median PFS 8.9 vs 1.3 yr vs chlorambucil (~10-yr). A041202 (≥65): 48-mo PFS 76% vs 47% (BR), rituximab adds nothing (HR 1.00), no OS gain; AF 18%, HTN 55%. E1912 (≤70): IR beat FCR, PFS HR 0.37, OS HR 0.47 — incl IGHV-mutated at 5.8 yr
Pirtobrutinib · BRUIN CLL-313	No del(17p), vs BR: PFS HR 0.20 , 24-mo PFS 93.4 vs 70.7%. FDA 1L approval 2 Oct 2026 (no del17p). CLL-314 vs ibrutinib: ORR noninferior (87.0 vs 78.5%), PFS still immature
VenO ×12 · CLL14	Unfit (CIRS >6 or CrCl <70), vs Clb-O: 6-yr median PFS 76.2 vs 36.4 mo , HR 0.40; OS HR 0.69 (P=.052, NS). Shorter PFS with del(17p) , IGHV-U, nodes ≥5 cm
CLL13 / GAIA fit, no TP53	GV / GIV / RV vs FCR-BR: 5-yr PFS 69.8 / 81.3 / 57.4 vs 50.7% ; month-15 uMRD GV 86.5% vs 52.0%. Venetoclax-rituximab was not better than CIT — obinutuzumab is the 1L partner. No OS difference
AV ± O · AMPLIFY	Fit, no del17p/TP53, vs FCR/BR: 36-mo PFS AV 76.5% / AVO 83.1% / 66.5% ; AV HR 0.65. AVO had more COVID deaths (25 vs 10) and lower 36-mo OS (87.7 vs 94.1%). FDA Feb 2026 for AV (14 cycles)
Ibrutinib + venetoclax · GLOW, CAPTIVATE	GLOW (older/unfit, vs Clb-O): 66-mo PFS 51.7 vs 18.1%, OS HR 0.46 . CAPTIVATE (phase 2, ≤70): 24-mo PFS 95%; after uMRD, placebo vs continued ibrutinib 1-yr DFS 95 vs 100% (NS); TP53 36-mo PFS 81%. EU-approved, not FDA-labeled
FLAIR MRD-guided I+V	Fit ≤75: I+V for 2× the time to uMRD (max 6 yr) vs FCR: 5-yr PFS 93.9 vs 58.1% , OS HR 0.26 ; vs ibrutinib alone PFS HR 0.29, OS HR 0.41. Cardiac SAEs 10.7% vs 0.4% (FCR)
CLL17 fixed vs continuous	VenO vs I+V vs continuous ibrutinib (n=909): 3-yr PFS 81.1 / 79.4 / 81.0% — both fixed-duration arms noninferior . PB uMRD 73.3 / 47.2 / 0%
FCR now niche	MDACC 12.8-yr PFS 53.9% if IGHV-mutated (plateau) vs 8.7% unmutated. Beaten by E1912, FLAIR, CLL13, AMPLIFY; myelosuppression, infections, therapy-related myeloid neoplasms. Never for TP53 or IGHV-U

Fixed vs continuous: CLL17 shows equal 3-yr PFS. Fixed duration = drug holiday, less cumulative cardiac/bleeding toxicity, needs TLS ramp-up and infusions (VenO). Continuous = simplest, preferred for TP53, but indefinite cost and AF/HTN. **Obinutuzumab, not rituximab**, with 1L venetoclax.

CHOOSING FIRST LINE — PATIENT FACTORS

Situation	Lean toward
del(17p) / TP53 mutation	Continuous zanubrutinib or acalabrutinib ; VenO if a BTKi is contraindicated (expect shorter PFS)
IGHV-mutated, no TP53	Fixed duration (VenO or AV) — longest treatment-free intervals; FCR no longer needed
IGHV-unmutated, no TP53	Either: fixed duration (VenO, AV, I+V) or continuous BTKi — CLL17 equal PFS; MRD-guided I+V (FLAIR) for fit
AF, heart failure, anticoagulation, bleeding risk	VenO ; if a BTKi is needed, zanubrutinib or acalabrutinib, never ibrutinib first
Renal impairment, bulky high-TLS-risk, can't do TLS labs	Continuous BTKi , or a BTKi lead-in before venetoclax (AV, I+V)
Frail / comorbid	Acalabrutinib, zanubrutinib or VenO (CLL14 enrolled CIRS >6)

RESPONSE AND MRD

- **iwCLL CR**: ALC <4×10⁹/L, nodes <1.5 cm, no hepatosplenomegaly, no B symptoms, ANC ≥1.5, plt ≥100, Hb ≥11 untransfused. Most patients on a BTKi reach PR, not CR — still durable
- **BTKi redistribution lymphocytosis** in the first weeks (cells leave nodes) is expected, **not progression**
- **uMRD = <10⁻⁴** (1 CLL cell in 10,000) by flow or NGS. Predicts PFS after fixed-duration therapy (MURANO, GLOW, CLL14); **meaningless on continuous BTKi** (CLL17 0%). MRD-guided duration is a trial/FLAIR concept, not routine

RELAPSED / REFRACTORY — EVIDENCE AND SEQUENCING

Regimen · trial	Result · caveat
VenR ×2 yr · MURANO	vs BR: median PFS 54.7 vs 17.0 mo ; 7-yr OS 69.6 vs 51.0%. uMRD at end of therapy predicts long PFS. VenR retreatment: ORR 72%, median PFS 23 mo. Mostly BTKi-naïve patients
Acalabrutinib vs ibrutinib · ELEVATE-RR	del17p/del11q: median PFS 38.4 mo both (noninferior); AF 9.4 vs 16.0%, HTN 9.4 vs 23.2%, any bleeding 38 vs 51%
Zanubrutinib vs ibrutinib · ALPINE	PFS superior, HR 0.65 (24-mo 78.4 vs 65.9%), incl del17p/TP53 (HR 0.53); AF 2.5 vs 10.1% (interim); fewer fatal cardiac events
Pirtobrutinib · BRUIN CLL-321	All post-cBTKi, vs idelalisib-R/BR: median PFS 14.0 vs 8.7 mo , HR 0.54; OS no different (crossover). Accelerated 2023 → regular approval 3 Dec 2025 : R/R after a cBTKi (BCL2i no longer required)
Pirtobrutinib-VR · BRUIN CLL-322	Fixed duration vs VR (80% cBTKi-exposed): 24-mo PFS 87 vs 72% , HR 0.55. Not FDA-approved as of Oct 2026
Liso-cel · TRANSCEND CLL 004	After BTKi + venetoclax: CR/CRi 18% , ORR 43%, blood uMRD 63%; median PFS 11.9 mo. CRS 85% (gr 3 9%), neurotox 45% (gr 3 18%). Accelerated approval Mar 2024 , ≥2 lines incl BTKi + BCL2i
Allo-HCT	Only curative option; for fit patients with double-exposed/refractory disease or Richter in remission
Not approved in CLL	BTK degraders (BGB-16673, bexobrutideg) and sonrotoclax — trials only (sonrotoclax is FDA-approved for R/R MCL)

RESISTANCE AND SEQUENCING RULES

- **Ibrutinib progression: BTK C481 or PLCG2 mutations in ~85%**, detectable a median 9.3 mo before clinical relapse. Covalent BTKis share C481 binding → **don't swap cBTKi for progression; pirtobrutinib (noncovalent) binds C481-mutant BTK**
- On pirtobrutinib, **non-C481 BTK mutations (T474I, L528W)** emerge; L528W and T474I also cross-resist zanubrutinib/acalabrutinib. Venetoclax resistance: **BCL2 G101V**
- **Intolerance ≠ resistance**: ibrutinib AF/bleeding → switch to acalabrutinib or zanubrutinib
- **Keep the BTKi going until the next therapy starts** — stopping can cause a rapid disease flare
- **Retest FISH + TP53 at every relapse**; new adenopathy with a high LDH → think Richter

RICHTER TRANSFORMATION

- **~2–10% of CLL (~0.5–1%/yr)**. Suspect with **rapid, asymmetric node growth, rising LDH, B symptoms, hypercalcemia**, new extranodal disease
- **PET-CT → biopsy the hottest lesion**. SUVmax ≥10 flags Richter (OS 6.9 vs 56.7 mo); **core or excisional, not FNA** (FNA often nondiagnostic)
- **Clonally related** (same IGHV as the CLL; more TP53 disruption) median OS **14.2 mo vs unrelated 62.5 mo** — unrelated behaves like de novo DLBCL
- **Treatment**: R-CHOP-type induction → **allo-HCT in responders**; CD19 CAR-T or bispecifics on trial/off-label. Hodgkin type: ABVD-type. No Richter-specific FDA approval. Median OS overall ~10 mo

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REGIMEN DOSING

Regimen	Dosing
Acalabrutinib	100 mg BID, continuous; A-O: obinutuzumab C2–7. Tablets OK with PPIs (old capsules: avoid)
Zanubrutinib	160 mg BID or 320 mg daily, continuous
Ibrutinib	420 mg daily, continuous
Pirtobrutinib	200 mg daily, continuous
Venetoclax ramp-up	20 → 50 → 100 → 200 → 400 mg, one step per week (5 weeks), with TLS prophylaxis (table)
VenO <i>CLL14, 12 × 28-d cycles</i>	Obinutuzumab 100 mg C1D1, 900 mg C1D2, 1000 mg C1D8 + D15, then 1000 mg D1 C2–6. Venetoclax ramp starts C1D22, 400 mg from C3, stop after C12
AV <i>AMPLIFY, 14 cycles</i>	Acalabrutinib 100 mg BID C1–14; venetoclax ramp-up from C3, 400 mg through C14. (AVO adds obinutuzumab C2–7 — not in US label)
I+V <i>GLOW / CAPTIVATE</i>	Ibrutinib 420 mg ×3 cycles lead-in (debulks: high-TLS-risk 21% → 1%), then + venetoclax ramp to 400 mg ×12 cycles (15 total)
VenR <i>MURANO</i>	Ramp-up 5 wk, then rituximab 375 mg/m² C1, 500 mg/m² C2–6 (q28d); venetoclax 400 mg for 24 mo from rituximab C1D1
Liso-cel	Flu/Cy lymphodepletion → 90–110×10⁶ CAR+ T cells ; REMS removed; stay near center, no driving ~2 wk

VENETOCLAX TLS RISK (LABEL) — REASSESS AFTER DEBULKING

Risk	Criteria → prophylaxis · monitoring
Low	All nodes <5 cm and ALC <25 → oral fluids 1.5–2 L/d + allopurinol (start 2–3 d before) · outpatient; labs pre, 6–8 h, 24 h at the first 20 and 50 mg doses
Medium	Any node 5 to <10 cm or ALC ≥25 → oral ± IV fluids + allopurinol · same labs; consider admission if CrCl <80
High	Any node ≥10 cm, or ALC ≥25 + any node ≥5 cm → oral + IV fluids 150–200 mL/h , allopurinol, rasburicase if uric acid high · admit for 20 and 50 mg ; labs 0, 4, 8, 12, 24 h

Labs = K, phos, Ca, uric acid, creatinine — **correct before the next dose**. Debulk first (BTKi lead-in in GLOW/AMPLIFY; obinutuzumab in VenO). **No urine alkalinization**.

FIXED-DURATION REGIMENS AT A GLANCE

Regimen	Length · route · uMRD · setting
VenO	12 cycles (~1 yr) · IV obinutuzumab C1–6 · PB uMRD after therapy 73.3% (CLL17) · 1L, any fitness
AV	14 cycles · all oral · 1L (AMPLIFY excluded del17p/TP53); FDA Feb 2026
I+V	15 cycles · all oral · PB uMRD 47.2% (CLL17) · 1L; not US-labeled
VenR	24 mo · rituximab C1–6 · R/R (MURANO)

All need the 5-week venetoclax ramp-up and TLS precautions; BTKi lead-in (AV, I+V) lowers TLS risk category.

BTK INHIBITORS COMPARED

Agent	Binding · signature toxicity
Ibrutinib	Covalent (C481), 1st gen · AF (16.0% in ELEVATE-RR), HTN , ventricular arrhythmia, bleeding, arthralgia
Acalabrutinib	Covalent, more selective · headache early; AF 9.4% vs 16.0%, HTN 9.4 vs 23.2% (vs ibrutinib)
Zanubrutinib	Covalent, more selective · neutropenia ; AF 2.5 vs 10.1% vs ibrutinib (ALPINE interim)
Pirtobrutinib	Noncovalent · works with C481 mutations; AF/flutter 3.1% (label, pooled); infections, rash

BTKI TOXICITY MANAGEMENT

- **Atrial fibrillation**: rate control (**avoid diltiazem/verapamil** — CYP3A), DOAC if indicated; often switch to a 2nd-gen BTKi or to venetoclax
- **Ventricular arrhythmia / sudden death**: ibrutinib label warning; FLAIR excess cardiac deaths. Palpitations/syncope → hold + ECG/Holter; don't rechallenge ibrutinib
- **Hypertension**: 55% all-grade on ibrutinib (A041202) — check BP every visit, treat early
- **Bleeding**: **no warfarin**; avoid fish oil/vitamin E; caution with DOAC + antiplatelet. **Hold 3–7 days before and after procedures**
- **Headache with acalabrutinib** (31–40%, early, self-limited): acetaminophen + caffeine. Also arthralgia, rash, diarrhea, bruising
- **CYP3A**: acalabrutinib — avoid strong inhibitors, 100 mg daily with moderate · zanubrutinib — 80 mg daily (strong), 80 mg BID (moderate) · pirtobrutinib — minus 50 mg with strong · **venetoclax with posaconazole 70 mg**, other strong 100 mg, moderate ≥50% cut; strong inhibitors contraindicated during ramp-up

COMPLICATIONS OF THE DISEASE

- **Autoimmune cytopenias ~5–10%: AIHA most common** (warm IgG), ITP ~2%, **PRCA** and autoimmune granulocytopenia <1%. Distinguish from marrow failure (retic, haptoglobin, DAT, marrow). **Steroids ± rituximab**; IVIG for ITP bleeding. **Steroid-refractory autoimmune cytopenia = iwCLL indication** — treat the CLL (BTKi or VenO; **avoid fludarabine monotherapy**)
- **Infection is the leading cause of death**: hypogammaglobulinemia + T-cell and neutrophil dysfunction. **IVIG if IgG <500 mg/dL + recurrent serious bacterial infections** (fewer infections, no survival gain)
- **Vaccines**: before therapy when possible; **no live vaccines; recombinant zoster** (2 doses), pneumococcal conjugate, influenza, COVID, RSV — responses blunted on BTKi/anti-CD20
- **Prophylaxis: anti-HBc+** → **entecavir** with anti-CD20 (HBV reactivation). **PJP + HSV/VZV** with FCR/BR, idelalisib (label) and after CAR-T; individualize on BTKi/venetoclax
- **Second cancers: skin SCC, BCC, melanoma, Merkel** markedly increased → **annual skin exam**, sun protection. Acalabrutinib label: second primaries 16% (non-melanoma skin 9%)

KEY TRIALS

SEQUOIA	Zanubrutinib vs BR 1L — PFS HR 0.29; del17p arm C works
ELEVATE-TN	Acalabrutinib ± O vs Clb-O — 6-yr PFS 78/62/17%
RESONATE-2	Ibrutinib 1L — median PFS 8.9 yr
E1912	IR vs FCR (≤70) — OS HR 0.47
CLL14	VenO ×12 vs Clb-O — median PFS 76.2 vs 36.4 mo
CLL13	GV/GIV beat CIT; RV did not
AMPLIFY	AV vs FCR/BR — PFS HR 0.65; FDA 2026
GLOW	I+V vs Clb-O — OS HR 0.46 at 67 mo
FLAIR	MRD-guided I+V vs FCR — OS HR 0.26
CLL17	Fixed duration noninferior to continuous
BRUIN CLL-313	Pirtobrutinib vs BR 1L — PFS HR 0.20
MURANO	VenR vs BR R/R — OS benefit at 7 yr
ALPINE · ELEVATE-RR	2nd-gen BTKi ≥ ibrutinib, less AF
BRUIN CLL-321	Pirtobrutinib post-cBTKi — PFS HR 0.54
TRANSCEND CLL 004	Liso-cel double-exposed — CR 18%
CLL12	Early ibrutinib: EFS only, no OS — watch and wait

ON-TREATMENT MONITORING

- **Venetoclax neutropenia** (gr 3–4 52.8% in CLL14): **G-CSF**, hold/dose-reduce per label rather than stopping; check CBC each cycle
- **Obinutuzumab infusion reactions** (40% with Clb-O vs 13% when given with acalabrutinib in ELEVATE-TN): split first dose 100/900 mg, premedicate (steroid, acetaminophen, antihistamine), hold antihypertensives the morning of dose 1
- **BTKi**: CBC, BP, pulse every visit; ask about palpitations, bruising, infections. Long-term: **~20% of ibrutinib patients stop for toxicity** (E1912: 21.9% by 5.8 yr)
- **Off therapy** (fixed duration or watch-and-wait): CBC + exam q3–6 mo; **imaging only for symptoms or before re-treatment**